

HOLZDORF, GermanyWMO: **104760**Lat: **51.7658N**Lon: **13.1667E**Elev: **333**StdP: **14.52**Time Zone: **1.00 (EUC)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	15.9	5.2	7.3	13.3	11.1	9.8	17.6	25.5	44.8	23.0	44.4	4.5	100	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	89.5	67.6	85.7	66.1	82.1	64.8	69.7	83.6	68.0	80.9	66.4	78.1	8.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	96.3	72.4	64.0	90.4	71.6	62.3	85.0	70.8	33.9	83.4	32.5	80.7	31.2	78.1	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.0	15.7	2.2	95.6	7.2	3.9	-2.9	98.4	-7.1	100.7	-11.1	102.8	-16.4	105.7		
(5)			1.9	73.0	7.1	1.8	-3.3	74.3	-7.5	75.4	-11.5	76.4	-16.6	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	32.7	34.5	40.2	48.9	57.1	63.0	66.8	65.6	58.3	49.7	40.9	35.4
	DBStd	14.01	9.51	8.23	7.23	7.29	6.95	6.39	6.02	5.70	5.94	6.67	6.95	7.93
	HDD50	2235	538	436	309	106	16	1	0	0	4	90	281	454
	HDD65	5924	1002	855	768	483	261	111	51	62	214	476	723	918
	CDD50	2054	1	1	5	74	236	392	520	483	254	79	8	1
	CDD65	270	0	0	0	1	16	53	106	80	14	0	0	0
	CDH74	3525	0	0	0	30	287	738	1271	978	215	6	0	0
	CDH80	1198	0	0	0	3	73	250	490	341	41	0	0	0
Wind	WSAvg	7.0	8.1	7.7	7.7	6.8	6.7	6.6	6.6	6.2	6.2	6.8	7.0	7.9
Precipitation	PrecAvg	22.1	1.7	1.3	1.5	1.2	2.2	2.1	2.9	2.4	1.9	1.4	1.8	1.7
	PrecMax	30.6	2.9	2.4	3.9	3.6	4.6	5.2	7.3	7.3	5.6	3.0	4.8	3.3
	PrecMin	14.0	0.2	0.3	0.3	0.0	0.2	0.5	0.8	0.6	0.4	0.3	0.0	0.5
	PrecStd	4.1	0.8	0.6	0.9	0.8	1.2	1.1	1.5	1.5	1.1	0.7	1.0	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.0	66.1	77.8	86.6	91.6	95.0	93.5	84.6	73.5	60.9	55.4
		MCWB	49.6	47.7	53.1	59.3	65.2	68.2	68.4	68.5	65.0	61.2	55.2	50.3
	2%	DB	51.5	52.8	60.8	72.5	81.4	86.4	89.8	88.0	80.2	67.8	56.9	51.7
		MCWB	47.1	46.7	49.9	56.6	62.6	66.2	67.5	67.5	63.4	58.3	51.7	47.6
	5%	DB	48.3	49.7	55.9	68.4	76.7	82.4	85.7	83.6	75.5	64.3	53.7	48.6
		MCWB	44.7	44.3	47.3	54.7	60.3	64.7	66.4	65.6	62.1	56.8	49.5	45.3
	10%	DB	45.0	46.4	52.1	64.4	72.4	77.4	81.0	79.1	71.2	60.6	51.3	46.3
		MCWB	42.2	42.5	46.0	52.7	57.9	62.5	65.1	64.1	59.9	55.0	47.7	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	49.4	54.0	60.3	67.4	71.6	71.8	71.4	67.9	62.5	55.8	51.3
		MCDB	54.2	54.4	63.4	74.8	82.6	85.5	87.4	87.4	79.5	70.7	60.0	54.7
	2%	WB	47.5	47.3	51.0	58.0	64.4	68.4	69.5	69.4	64.9	59.6	52.5	47.9
		MCDB	50.4	52.0	57.9	70.1	76.7	80.8	83.5	83.4	75.4	66.0	56.0	50.9
	5%	WB	44.7	44.8	48.8	55.8	62.0	66.1	67.9	67.5	63.0	57.5	50.1	45.7
		MCDB	47.7	48.6	55.0	66.3	73.5	78.9	81.2	79.5	72.6	63.2	52.8	48.4
	10%	WB	42.3	42.2	46.3	53.6	59.6	63.9	66.2	65.7	61.3	55.4	48.1	43.2
		MCDB	45.0	46.1	51.7	62.8	69.5	75.5	77.7	75.8	69.8	59.8	50.8	45.6
Mean Daily Temperature Range	5% DB	MDBR	10.4	12.8	16.4	21.9	22.6	22.1	21.6	22.6	20.4	16.2	11.7	9.5
		MCDBR	12.1	17.2	25.2	29.2	30.3	30.5	30.7	30.2	28.2	22.0	14.3	11.4
		MCWBR	9.2	11.8	15.5	15.6	14.0	12.9	11.8	12.9	13.9	13.6	10.1	8.7
	5% WB	MCDBR	11.3	15.3	22.5	26.6	26.3	27.1	25.4	25.3	24.0	20.2	13.1	10.7
		MCWBR	9.0	11.0	14.7	14.6	13.0	12.5	10.7	12.0	12.7	13.1	9.9	8.7
Clear-Sky Solar Irradiance	taub		0.320	0.345	0.387	0.417	0.418	0.414	0.433	0.426	0.388	0.369	0.344	0.314
	taud		2.341	2.317	2.215	2.194	2.236	2.275	2.230	2.266	2.346	2.364	2.374	2.354
	Ebn at Noon		215	243	252	259	265	267	259	254	251	230	204	197
	Edh at Noon		20	27	36	41	42	40	42	38	31	26	20	17
All-Sky Solar Radiation	RadAvg		223	441	761	1259	1547	1649	1554	1343	971	556	263	170
	RadStd		27	76	93	161	185	134	200	114	118	65	36	18